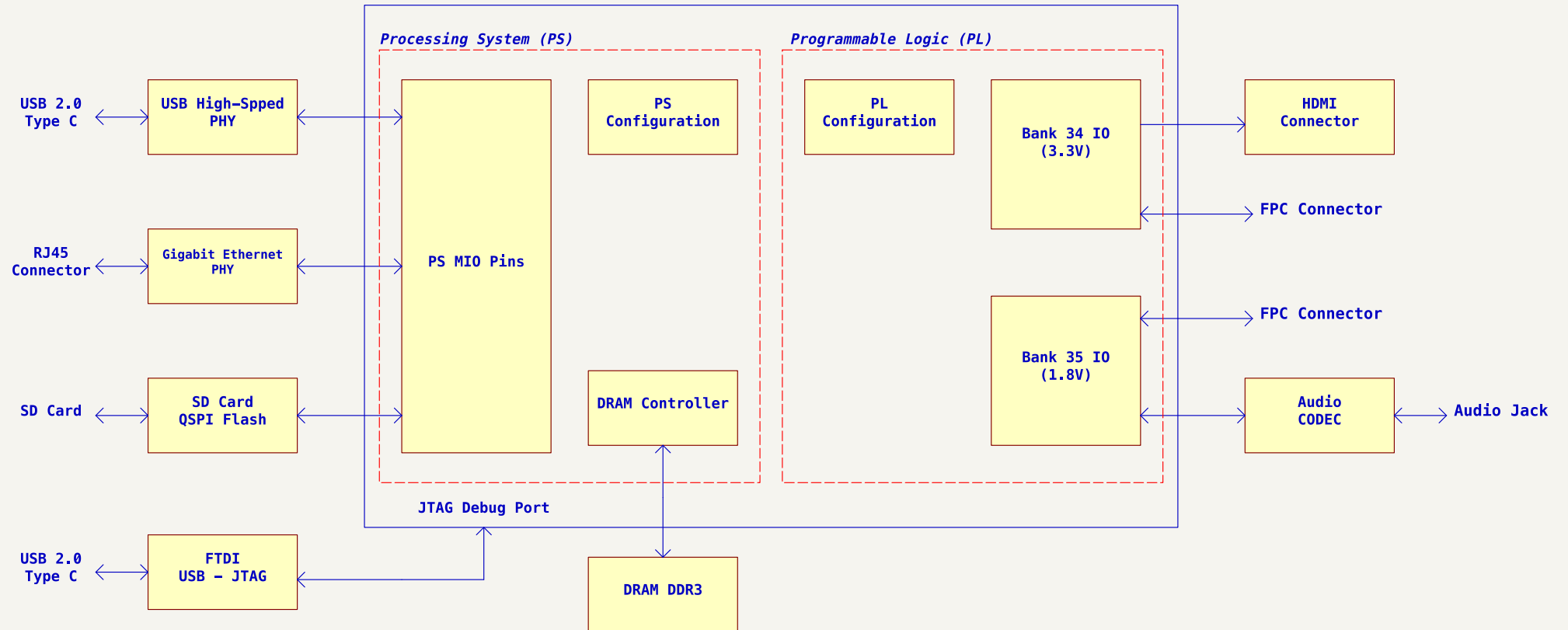
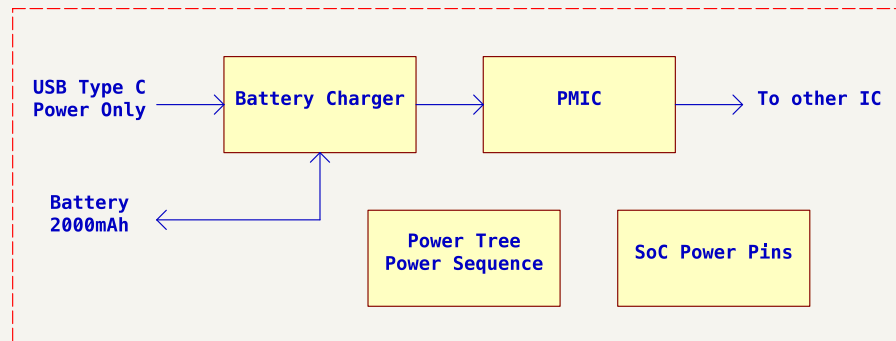


Overview: Zynq7000 Development Board

XC7Z010-1CLG400C SoC

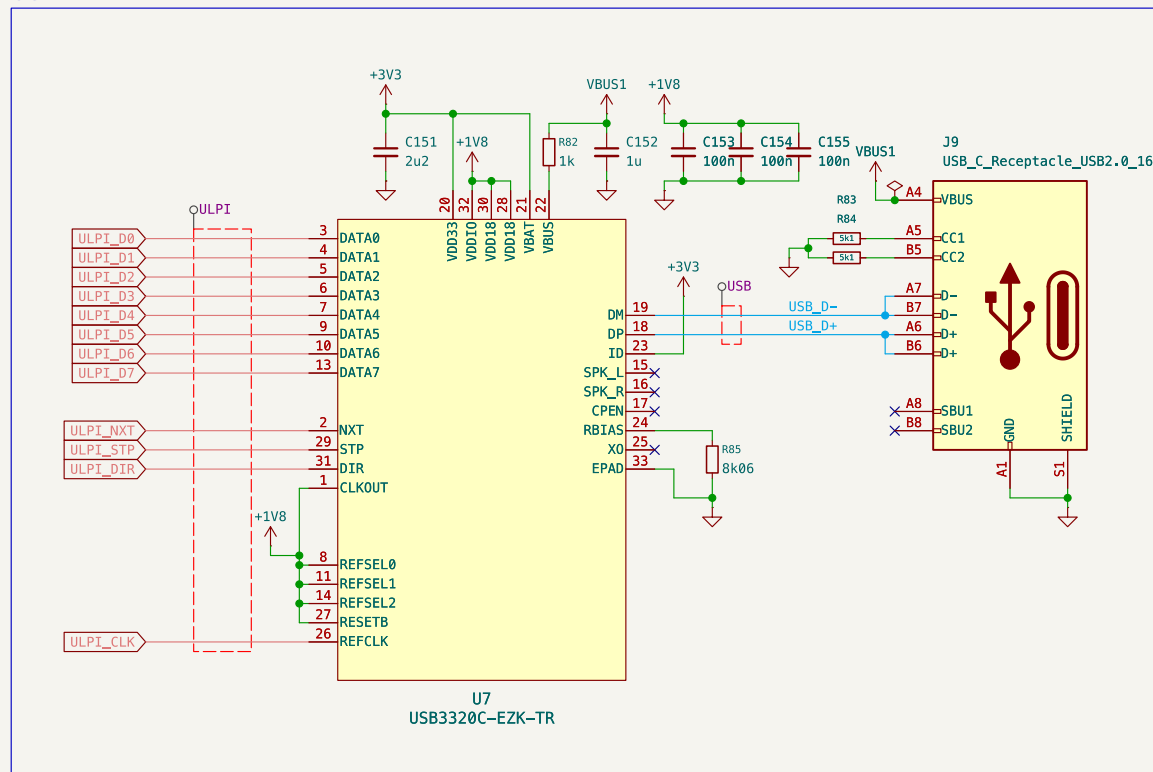


Power Block



Sheet: /		D
File: zynq7000.kicad_sch		
Title: FPGA Zynq7000 Dev Board		
Size: A4	Date: 2025-07-10	
KiCad E.D.A, 9,0,3		
Rev: A		
Id: 1/18		

USB PHY



Sheet: /USB HS PHY/
File: ps_usb.kicad_sch

Title: FPGA Zynq7000 Dev Board

Size: A4 Date: 2025-07-10

KiCad E.D.A. 9.0.3

Rev: A

Id: 13/18

The schematic diagram illustrates the connections for the RTL8211F-CG Ethernet PHY (U6A). The chip is shown with its pins and internal components, including resistors (R69, R70, R71, R72, R73, R74, R76, R77, R78, R79, R81) and LEDs (ETH_LED0, ETH_LED1, ETH_LED2). The connections are as follows:

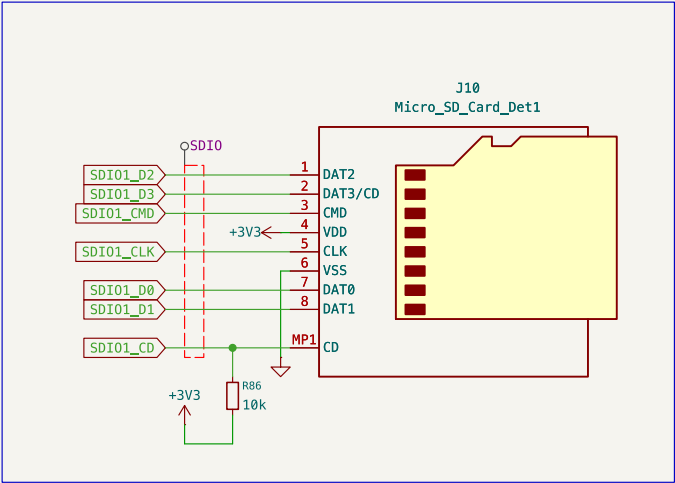
- ETH_RGMII (U6B):**
 - ETH_TXCK (20) to TXC
 - ETH_TXD0 (18) to TXD0
 - ETH_TXD1 (17) to TXD1
 - ETH_TXD2 (16) to TXD2
 - ETH_TXD3 (15) to TXD3
 - ETH_TXCT (19) to TXCTL
 - ETH_RXCK (27) to RXC/PHYAD1
 - ETH_RXD0 (25) to RXD0/RXDLY
 - ETH_RXD1 (24) to RXD1/TXDLY
 - ETH_RXD2 (23) to RXD2/PLLOFF
 - ETH_RXD3 (22) to RXD3/PHYAD0
 - ETH_RXCT (26) to RXCTL/PHYAD2
- ETH_MIO (U6B):**
 - ETH_MIO (13) to MDC
 - ETH_MIO (14) to MDIO
 - ETH_MIO (31) to INTB/PMEB
- ETH_XTAL (U6B):**
 - ETH_XTAL_IN (36) to XTAL_IN
 - ETH_XTAL_OUT (37) to XTAL_OUT/EXT_CLK
 - ETH_XTAL_OUT (35) to CLKOUT
- ETH_LED (U6B):**
 - ETH_LED0 (32) to LED0/CFG_EXT
 - ETH_LED1 (33) to LED1/CFG_LD00
 - ETH_LED2 (34) to LED2/CFG_LD01
 - ETH_Nrst (12) to PHYRSTB
- ETH_MDI (U6C):**
 - MDIN3 (10) to MDI3-
 - MDIP3 (9) to MDI3+
 - MDIN2 (7) to MDI2-
 - MDIP2 (6) to MDI2+
 - MDIN1 (5) to MDI1-
 - MDIP1 (4) to MDI1+
 - MDIN0 (2) to MDI0-
 - MDIP0 (1) to MDI0+
- Other Connections:**
 - R69 (22k) between ETH_TXCK and ETH_RXCK
 - R70 (22k) between ETH_TXD0 and ETH_RXD0
 - R71 (22k) between ETH_TXD1 and ETH_RXD1
 - R72 (22k) between ETH_TXD2 and ETH_RXD2
 - R73 (22k) between ETH_TXD3 and ETH_RXD3
 - R74 (22k) between ETH_TXCT and ETH_RXCT
 - R76 (1k) between ETH_MIO and ETH_XTAL_IN
 - R77 (1k) between ETH_LED0 and ETH_LED1
 - R78 (2k49) between ETH_LED1 and ETH_LED2
 - R79 (1k) between ETH_LED2 and ETH_Nrst
 - R81 (1k) between ETH_Nrst and ETH_XTAL_OUT

[illegible]

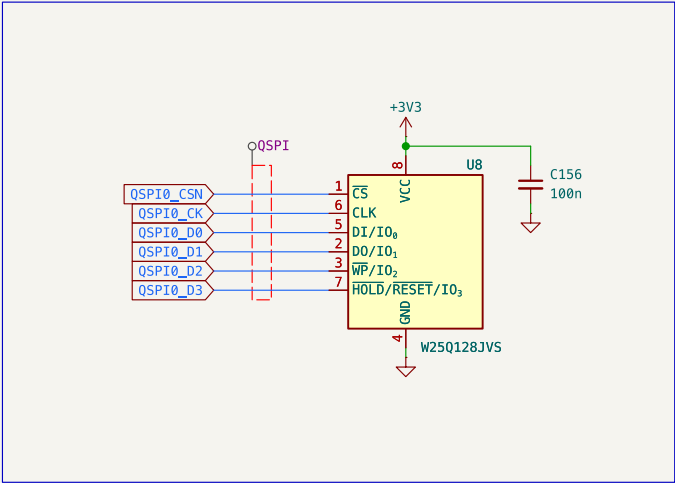
Size: A4	B
KiCad E.D.A. 9.0.3	

Id: 12/18

SD Card



QSPI Flash



Sheet: /SD Card + QSPI Flash/
File: ps_sd_qspi.kicad_sch

Title: FPGA Zynq7000 Dev Board

Size: A4

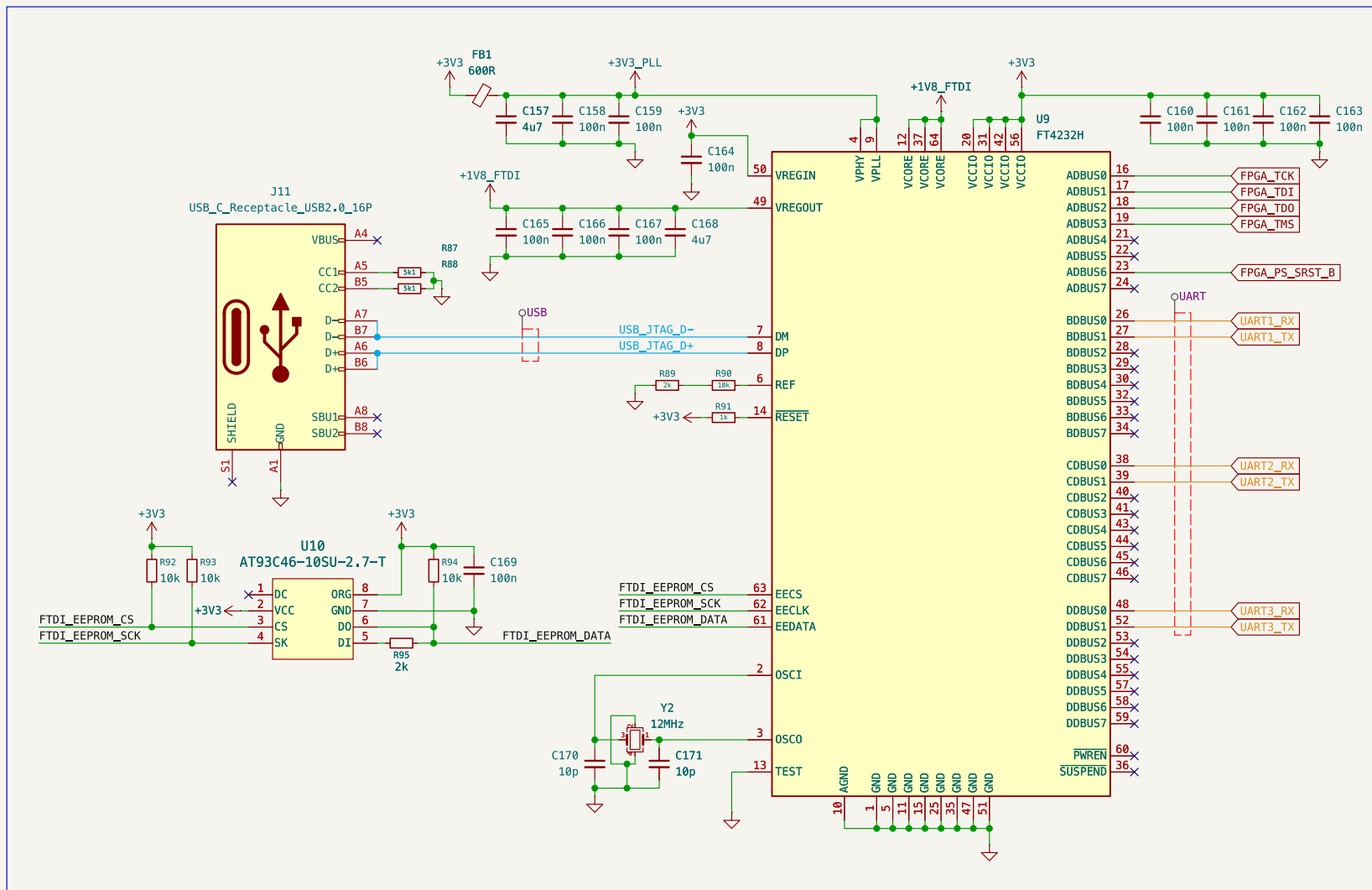
Date: 2025-07-10

Rev: A

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Id: 14/18

FTDI USB-JTAG



Sheet: /FTDI/

File: ftdi.kicad_sch

Title: FPGA Zynq7000 Dev Board

Size: A4

Date: 2025-07-10

Rev: A

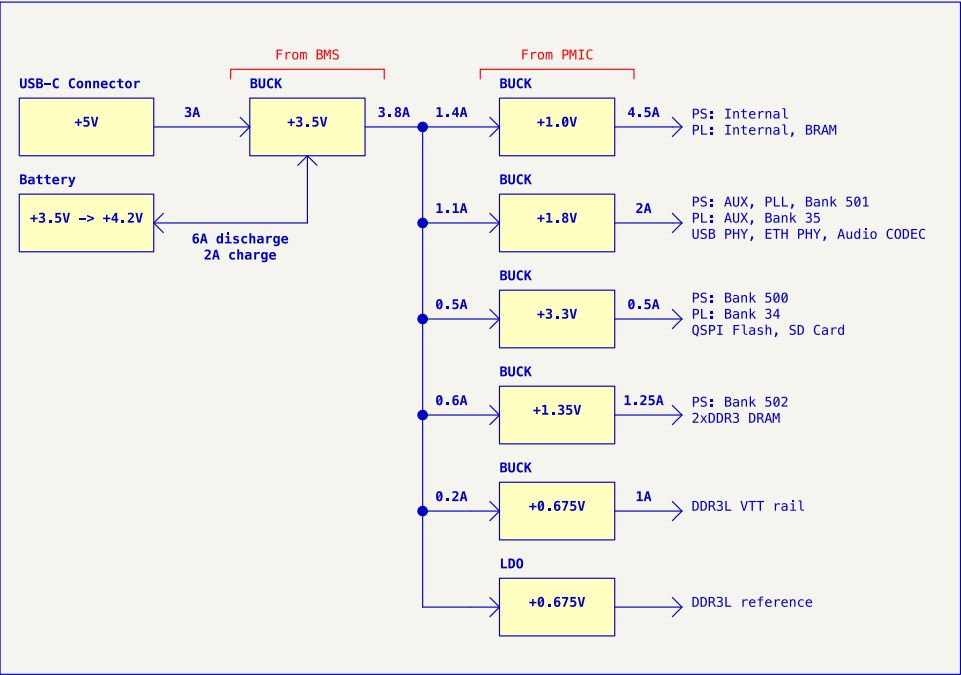
KiCad E.D.A. 9.0.3

Id: 15/18

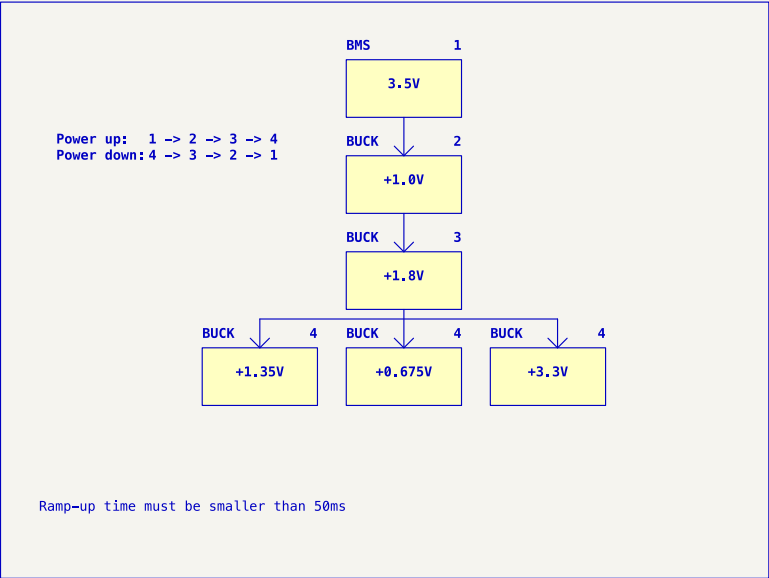
[illegible]

Id: 10/18

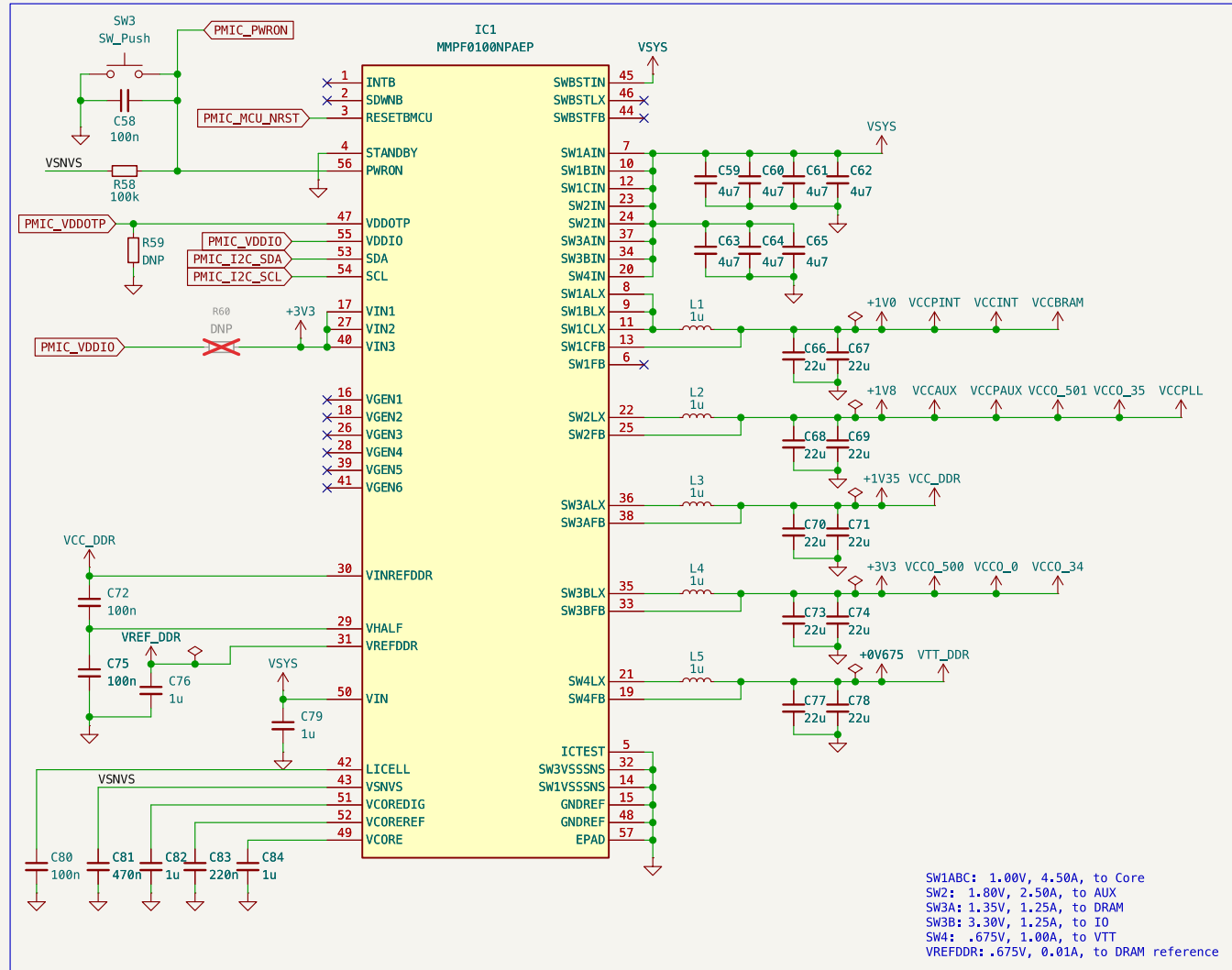
Power Tree



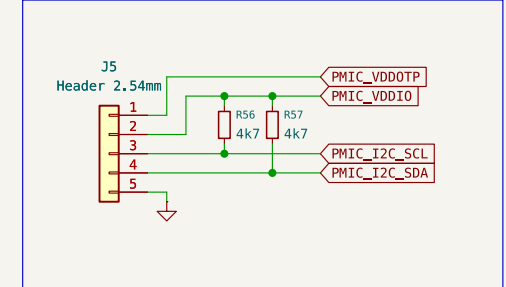
Power Sequence



PMIC



PMIC programming header



OK INTB, SDWNB: Leave floating
 OK RESETBMCU. Pull up to 3.3V (< Vin) with 100k.
 OK STANDBY, ICTEST = GND
 OK SW1ABC single phase mode
 OK SW1FB floating
 OK SW1CFB real feedback
 OK VIN123 connected to V < 3.4V
 OK LICELL bypass with 0.1u
 OK VSNVS bypass with 0.47u
 OK SWBSTFB floating
 OK SWBSTIN to VIN
 OK SWBSTLX floating
 N/A VDDOTP connected to VCOREDIG with 100k for default mode
 OK Connect to ground in fuse mode
 OK Connect to 8.0V to program
 OK VCORE 1u
 OK VIN 1u
 OK VCOREDIG 1u
 OK VCOREREF 0.22u
 OK SDA SCL pull to VDDIO
 OK VDDIO = 3.3V, Bypass 0.1u
 OK PWRON. Pull up to VSNVS
 RESETBMCU: In its default mode, it is de-asserted 2.0 ms to 4.0 ms after the last regulator in the start-up sequence is enabled;
 OK Power on: PWRON transitions from high to low
 OK Power off: PWRON is held low for longer than 4.0 seconds.

Sheet: /PMIC/
 File: pmic.kicad_sch

Title: FPGA Zynq7000 Dev Board

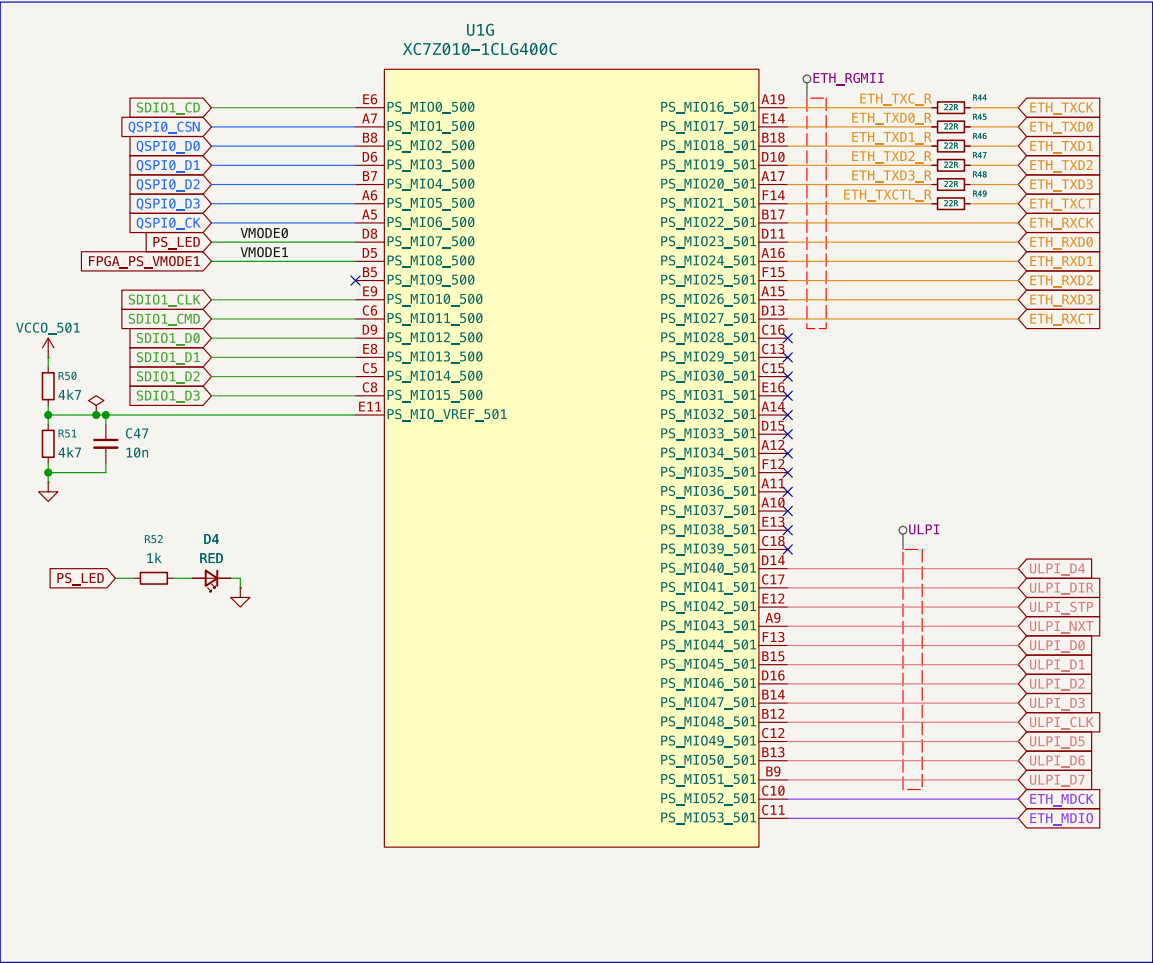
Size: A4 Date: 2025-07-10

KiCad E.D.A. 9.0.3

Rev: A

Id: 9/18

PS IO pins



The image displays four decoupling capacitor networks and a component footprint for the XC7Z010-1CLG400C.

Network 1: VCCPINT

- Capacitors: C92 (100u), C93 (4u7), C94 (470n), C95 (470n), C96 (470n)

Network 2: VCCQ_500 and VCCQ_501

- VCCQ_500:** C110 (100u), C111 (4u7)
- VCCQ_501:** C112 (470n), C113 (100u), C114 (4u7), C115 (470n)

Network 3: VCCQ_DDR

- Capacitors: C125 (4u7), C126 (470n), C127 (470n), C128 (470n), C129 (470n)

Component Footprint: U1H

Pin connections for U1H (XC7Z010-1CLG400C):

- Left Side:** F11, R12, R20, R8, T13, T3, T7, U16, U6, V19, V9, W12, W2, Y15, Y5, L8, M1, M11, M13, M7, N10, N12, N14, N4, N8, P11, P13, P17, P7, P9
- Right Side:** K11, K13, K15, K5, K7, L12, L18, C9, C4, C14, B11, H11, D17, E10, A18, A8, B1, E20, F3, F7, G10, G12, G16, H13, H19, H7, H9, J12, J2, J8

Power and Ground connections are indicated by red arrows at the top and bottom of the footprint.

The schematic diagram illustrates the power supply network for the AD9288, showing decoupling capacitors for various power planes. The components are as follows:

- VCCPAUX:** C107 (100u), C108 (4u7), C109 (470n).
- VCCINT:** C97 (100u), C98 (4u7), C99 (470n), C100 (470n).
- VCCBRAM:** C101 (47u), C102 (4u7).
- VCCAUX:** C103 (470n), C104 (47u), C105 (4u7), C106 (470n).
- VCC0_35:** C116 (47u), C117 (4u7), C118 (4u7), C119 (470n), C120 (470n), C121 (470n), C122 (470n).
- VCC0_34:** C130 (47u), C131 (4u7), C132 (4u7), C133 (470n), C134 (470n), C135 (470n), C136 (470n).
- VCC0_0:** C123 (47u).
- VCCPLL:** C124 (470n).

Size: A4 Date: 2025-07-10

Rev: A

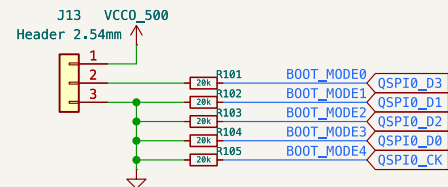
Id: 11/18

PS Configuration

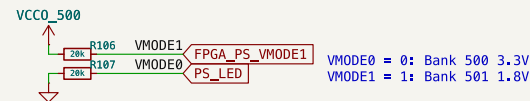
Table 6-4: Boot Mode MIO Strapping Pins

Pin-signal / Mode	MIO[8]	MIO[7]	MIO[6]	MIO[5]	MIO[4]	MIO[3]	MIO[2]
	VMODE[1]	VMODE[0]	BOOT_MODE[4]	BOOT_MODE[0]	BOOT_MODE[2]	BOOT_MODE[1]	BOOT_MODE[3]
Boot Devices							
JTAG Boot Mode; cascaded is most common ⁽¹⁾				0	0	0	JTAG Chain Routing ⁽²⁾ 0: Cascade mode 1: Independent mode
NOR Boot ⁽³⁾				0	0	1	
NAND				0	1	0	
Quad-SPI ⁽³⁾				1	0	0	
SD Card				1	1	0	
Mode for all 3 PLLs							
PLL Enabled			0	Hardware waits for PLL to lock, then executes BootROM.			
PLL Bypassed			1	Allows for a wide PS_CLK frequency range.			
MIO Bank Voltage ⁽⁴⁾							
	Bank 1	Bank 0	Voltage Bank 0 includes MIO pins 0 thru 15.				
2.5 V, 3.3 V	0	0	Voltage Bank 1 includes MIO pins 16 thru 53.				
1.8 V	1	1					

- Notes:
- JTAG cascaded mode is most common and is the assumed mode in all the references to JTAG mode except where noted.
 - For secure mode, JTAG is not enabled and MIO[2] is ignored.
 - The Quad-SPI and NOR boot modes support execute-in-place (this support is always non-secure)
 - Voltage Banks 0 and 1 must be set to the same value when an interface spans across these voltage banks. Examples include NOR, 16-bit NAND, and a wide TPIU test port. Other interface configuration may also span the two banks.

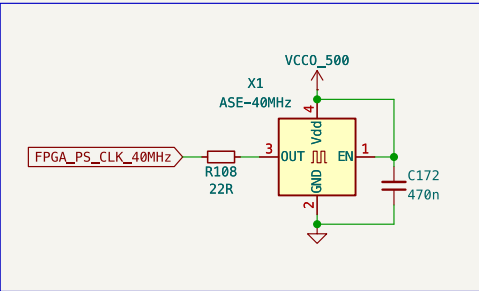


No NOR boot
No NAND boot
No SD Card boot
JTAG cascaded mode
QSPI boot or JTAG boot
PLL Enabled

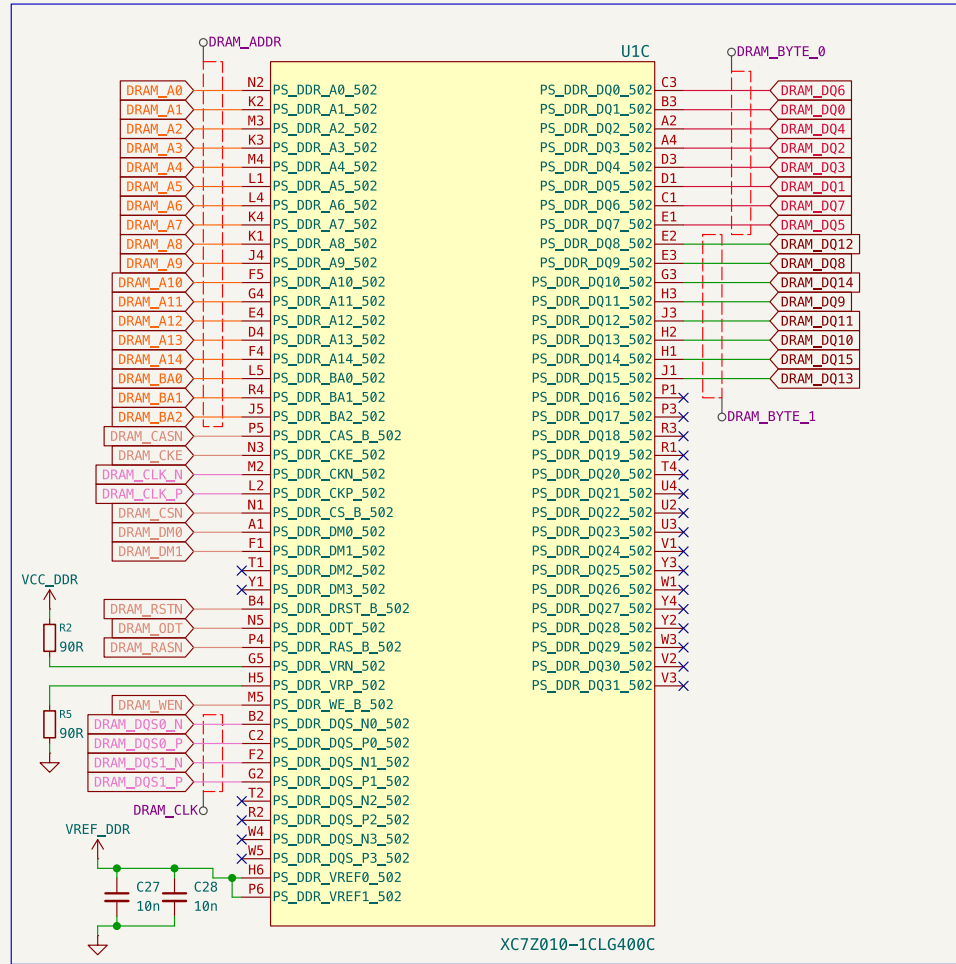


VMODE0 = 0: Bank 500 3.3V
VMODE1 = 1: Bank 501 1.8V

PS clock source



DRAM Controller



Sheet: /DRAM Controller/

File: ps_dram_ct.kicad_sch

Title: FPGA Zynq7000 Dev Board

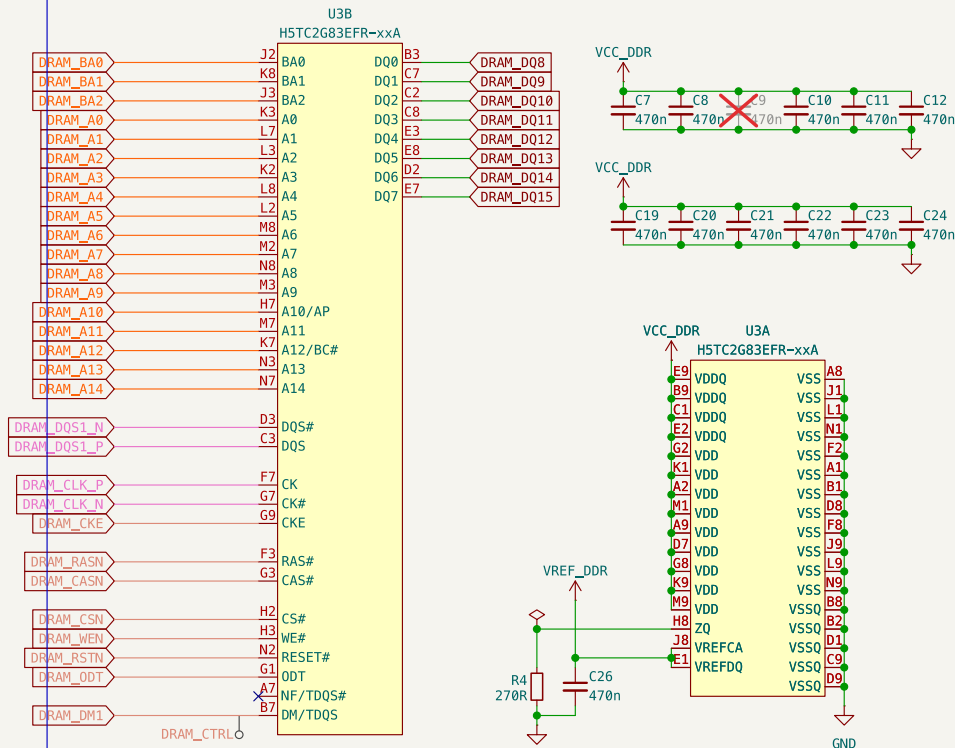
Size: A4 Date: 2025-07-10

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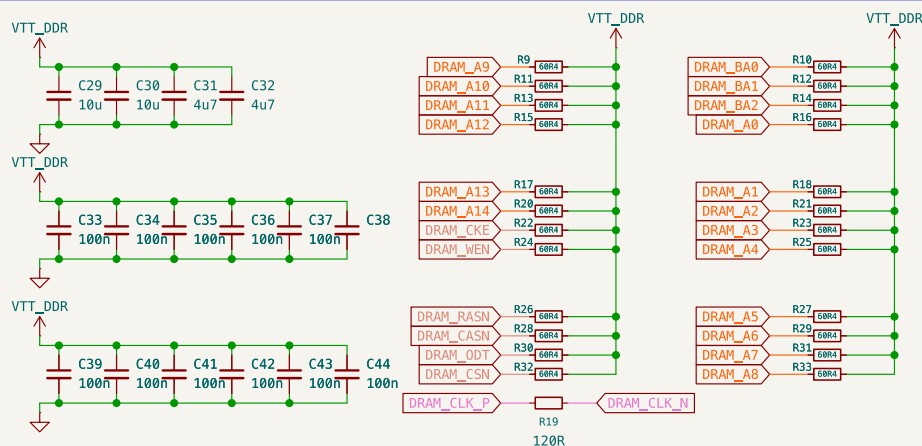
Rev: A

Id: 18/18

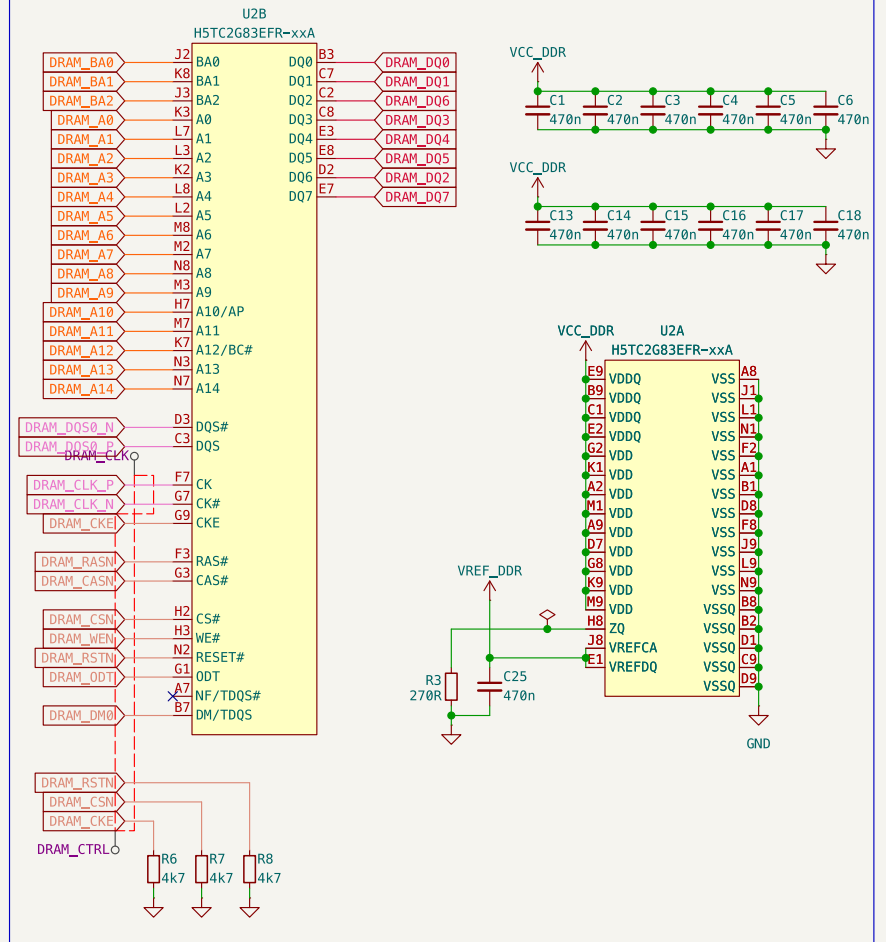
DDR3L-2



VTT bypass cap. and termination



DDR3L-1



Sheet: /DRAM DDR3/

File: ps_ddr3.kicad_sch

Title: FPGA Zynq7000 Dev Board

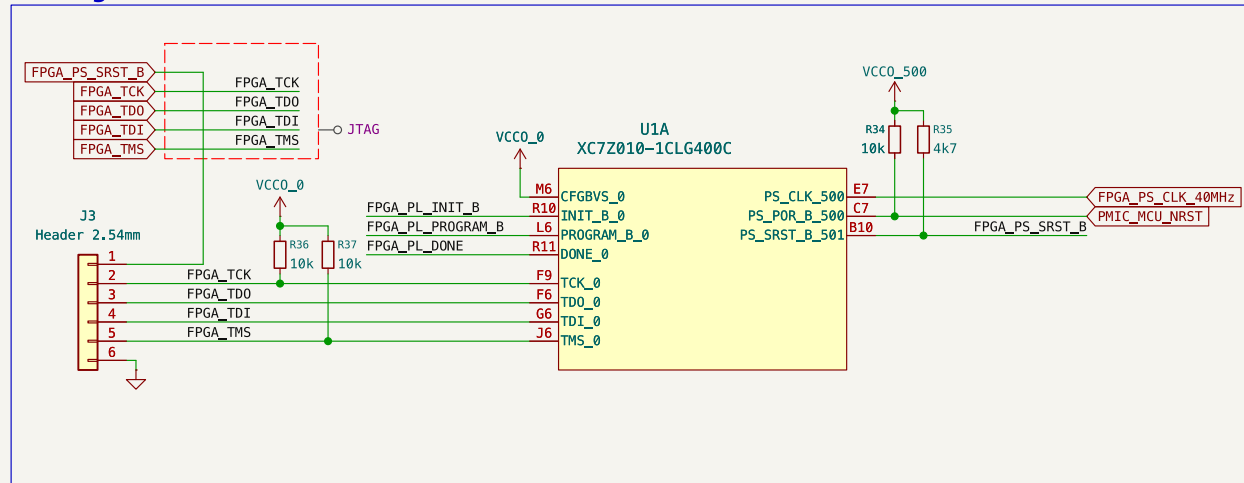
Size: A4 Date: 2025-07-10

KiCad E.D.A. 9.0.3

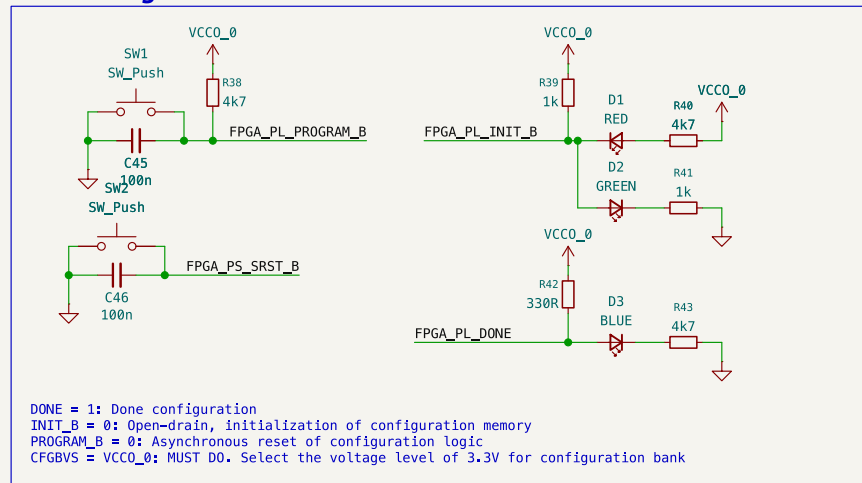
Rev: A

Id: 5/18

Configuration Block and JTAG conn.

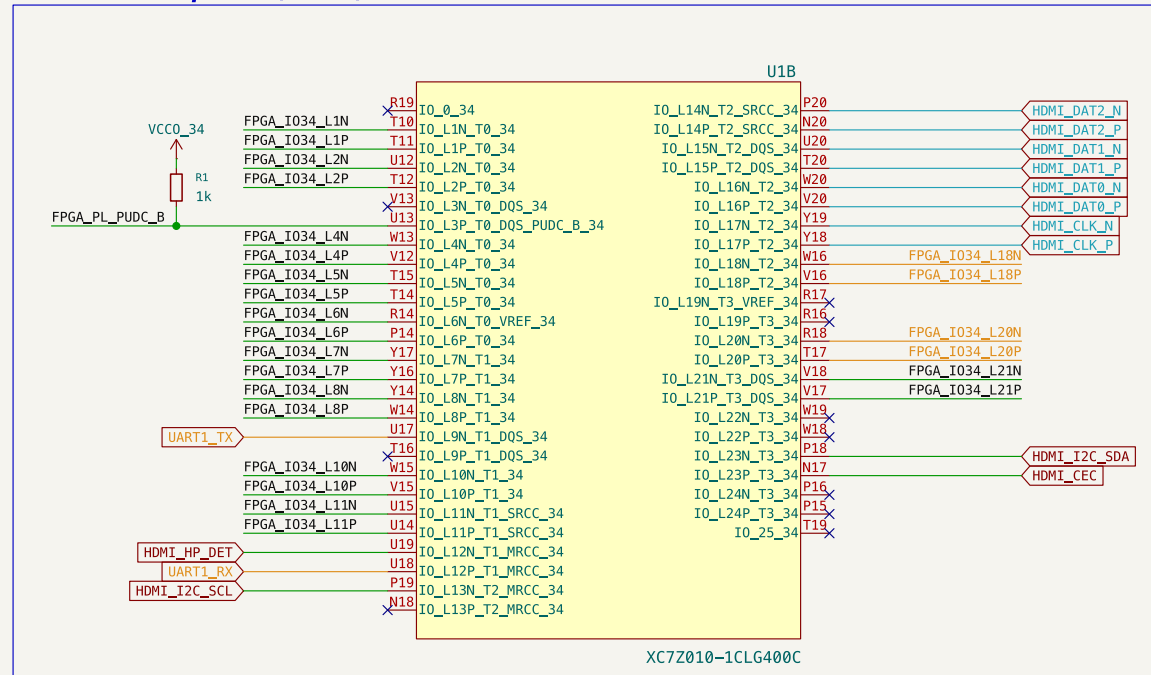


PL Configuration

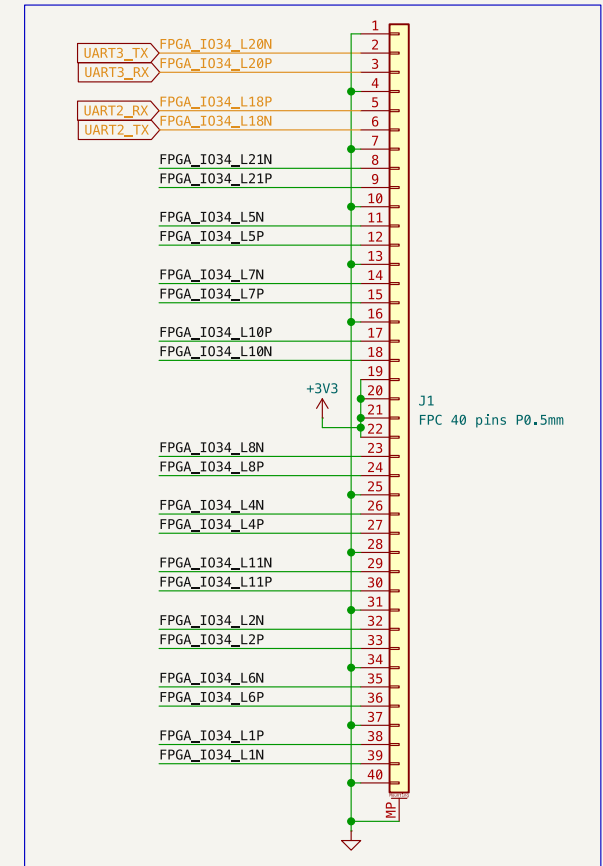


```
OK PS_POR_B input is required to be
asserted to GND during the power-on sequence
OK The PS system reset (PS_SRST_B) is an active-Low signal
that is mostly used for debugging proposes.
OK PS_SRST_B must be High to begin the boot process
OK 30MHz < PS_CLK < 60MHz
OK TMS and TCK have 10k pull up
OK PROGRAM_B pull up
OK INIT_B: 0 when init, error. 1 when done. pull up < 4k7
```

Bank 34 IO pins (3.3V)



Extended connector



Sheet: /Bank 34 (3.3V)/
File: pl_bank34.kicad_sch

Title: FPGA Zynq7000 Dev Board

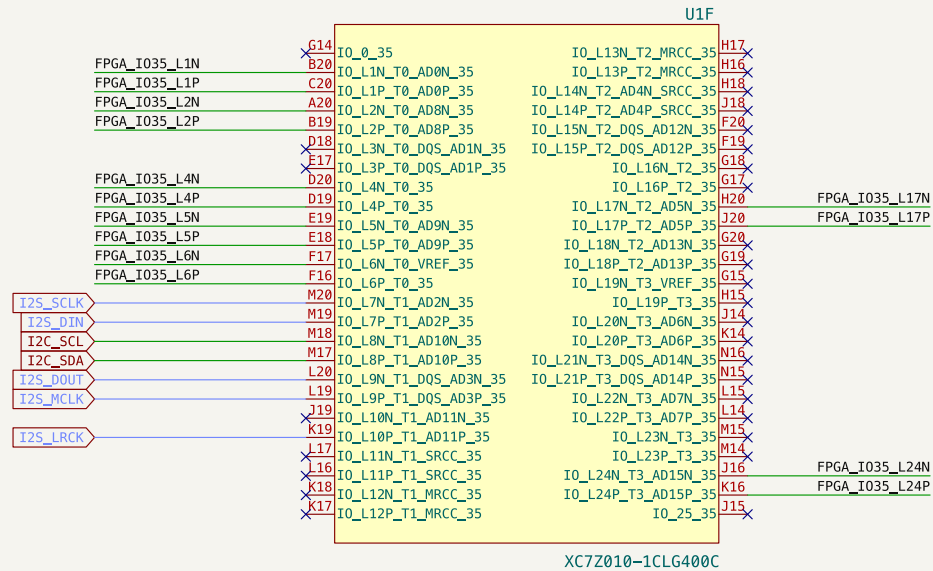
Size: A4 Date: 2025-07-10

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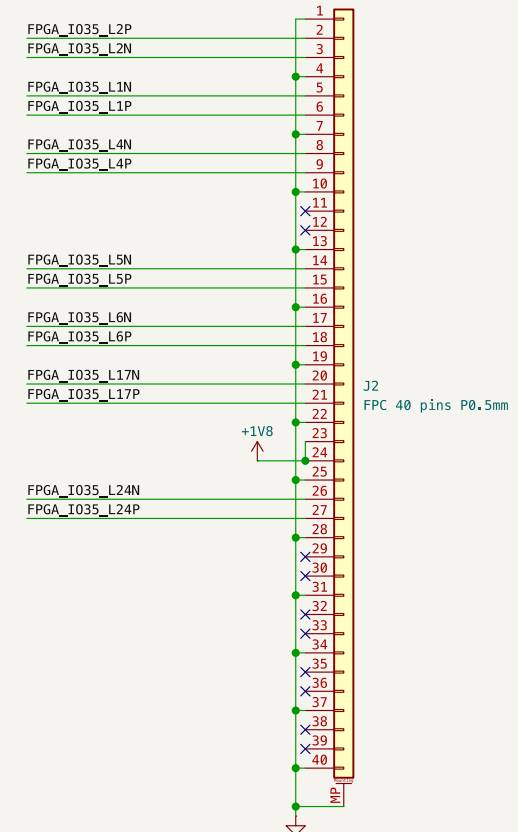
Rev: A

Id: 3/18

Bank 35 IO pins (1.8V)



Extended connector



Sheet: /Bank 35 (1.8V)/
File: pl_bank35.kicad_sch

Title: FPGA Zynq7000 Dev Board

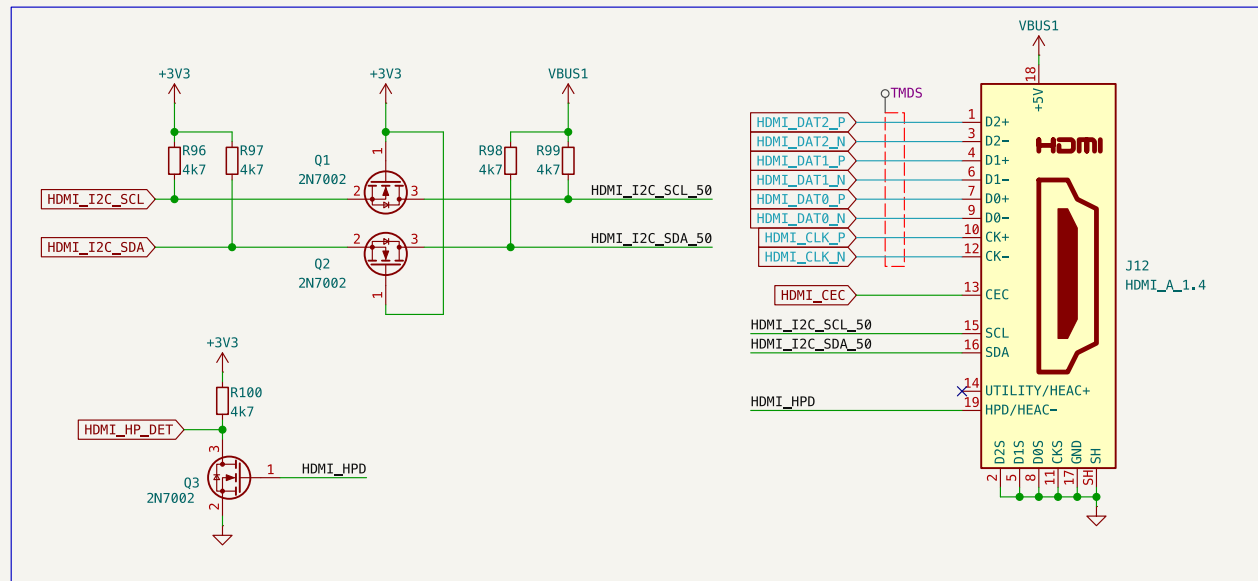
Size: A4 Date: 2025-07-10

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Rev: A

Id: 4/18

HDMI



Sheet: /HDMI/

File: pl_hdmi.kicad_sch

Title: FPGA Zynq7000 Dev Board

Size: A4

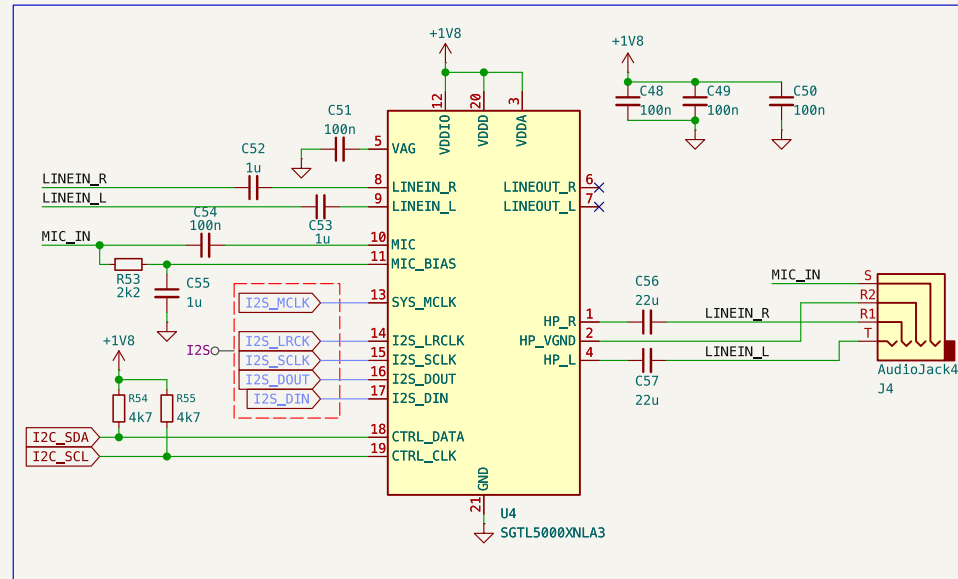
Date: 2025-07-10

Rev: A

KiCad E.D.A. 9.0.3

Id: 16/18

Audio CODEC



Sheet: /Audio CODEC/
File: pl_codec.kicad_sch

Title: FPGA Zynq7000 Dev Board

Size: A4 Date: 2025-07-10

KiCad E.D.A. 9.0.3

Rev: A

Id: 8/18